

Purification and Identification of an Unknown Solid

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1. EXPERIMENTAL METHOD

The Benzil(1.00 g) with four boiling chips was placed in a 50 mL beaker. The ethanol was heated up to boiling with a 150 mL beaker and four boiling chips using heating station (Set to around 325 degrees Fahrenheit). The boiled ethanol was poured into the 50 mL beaker containing the unpurified Benzil crystals and was placed on the heating station with thorough stirring to obtain a saturated Benzil solution.

When the unpurified Benzil was completely dissolved, the obtained saturated solution was poured into a funnel with filter paper. Obtained filtered solution was collected. After completing the filtration, the beaker with the filtered solution was left to cool down and then placed in ice water to accelerate the crystallization process. After crystallization, the mother solution with crystals was poured into a vacuum tube to separate the liquid from the crystals. If the green substance was visible on the surface of the purified Benzil yellow crystals, ice-cooled ethanol was used to clean off.

The unknown solid (1.00 g, NO.1) was purified using the same procedure given above. Obtained crystal NO.1 was dried before the melting point range measurement.

2. RESULT

Table I. Melting Point Ranges for the Impure and Purified Benzil

Purity	T_s	T_m
-	($^{\circ}C$)	($^{\circ}C$)
Pure	97.5	98.5
Impure	96.1	119

Table II. Melting Point Ranges for the Purified Compound and Two Mixtures

Tube	T_s	T_m
	($^{\circ}C$)	($^{\circ}C$)
1	59.5	62.8
2	57.5	61.2
3	50.1	74.8

In Table I, T_s is the temperature when solid starts to melt, T_m is the temperature when solid melted completely.

In Table II, T_s is the temperature when solid starts to melt, T_m is the temperature when solid melted completely. Tube 1 was filled by unknown solid (melting point range 59.5-62.8 $^{\circ}C$). Tube 2 (sample A+unknown) was filled with

the mixture with palmitic acid (melting point 63°C). Tube 3 (sample B + unknown) was filled with the mixture with phenylacetic acid (melting point 77°C).

3. DISCUSSION

3.1. *Melting Point Ranges for the Impure and Purified Benzil*

Table 1 shows that impure benzil has a wider melting temperature range ($96.1\text{--}118.5^{\circ}\text{C}$) and a lower melting starting point ($T_s = 96.1^{\circ}\text{C}$) compared to pure benzil, which is $97.5\text{--}98.5^{\circ}\text{C}$.

The melting point range was impacted by the purity of the benzil; the higher purification of the benzil was, the narrower the melting point range. The impurified mixed benzil was observed with a lower melting start point, and broader melting point range since the mixture of impurities lower the melting point of the substance.

3.2. *Melting Point Ranges for the Purified Compound and Two Mixtures*

Three tubes were selected: the target solid tube 1 (labeled as A), identical solid tube 2 (labeled with A+A), and mixed solid tube 3 (labeled as A+B).

Since adding a pure solid to an identical solid will not change the melting point range, the selected known solid can be considered as the identification reference. If the melting point range of the target solid did not change after mixing with another known solid, then the identity of the target solid could be determined given by the known solid since only purity mixed with purity does not change the melting point range. On the other hand, if the melting point range of the solid changed after mixing with another different type of solid, then the difference between the pure target solid and the mixing solid would be observed. Therefore, in this lab, the unknown solid's identity can be determined by comparing two mixtures of known solids: A and B.

The unknown target solid was filled to the melting point tube 1, the known solid A was filled into tube 2, and the known solid B was filled in tube 3. Three tubes would be placed in 3 wells in the same MEL-TEMP.

The melting point range of the unknown solid was measured before three tubes of simultaneous observation. The rough melting point range of the unknown is $59\text{--}62.8^{\circ}\text{C}$ by quick measurement. Therefore, two of the four options of the known solid were selected, with closest melting range point to the unknown.

Solid A is the palmitic acid (melting point 63°C) and the solid B is phenylacetic acid (melting point 77°C), given by the label of their container.

1. The target unknown without impurity was filled in the tube 1.
2. The target unknown mixed with the solid A was filled in the tube 2.
3. The target unknown solid mixed with the solid B was filled in tube 3.

Three tubes were placed in the three wells separately of MEL-TEMP.

When the preparation for observing the melting process was finished, the MEL-TEMP was turned on and set the power of 3 - 4 with the melting speed of 2°C per min. At the end of the melting point range observation, three measurements were collected. Table 2 shown that the ranges of tube 1 and 2 were identical, and the ranges of tube 1 and tube 3 were inconsistent. The identical melting point range of tube 1 and 2 indicated that the unknown target solid was palmitic acid since the range was almost consistent. However, the variation of ranges of tube 1 and 3 implied that the unknown target solid was NOT phenylacetic acid. Therefore, the identity of the unknown was determined by the comparison of the solid A, which was palmitic acid.

3.3. *Uncertainty and Error*

The purity of ethanol could be decreased due to its easy vaporation property. In the process of recrystallization, the purity would be decreased since the failure to separate the solid and the mother liquid. The mixture of the impurity

would be carried to the following procedure. Therefore, the target crystalization was polluted and the melting point range changed as a result.